

JISU NAM

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RESEARCH INTEREST

My research bridges visual perception and generation. I am interested in general perception for images and videos, particularly visual correspondence, such as image matching and point tracking. Recently, I have been working on large-scale image/video generative models, analyzing their representations in the context of correspondence, ultimately to understand, enhance, and manipulate the synthesis process.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) M.S./Ph.D. in Artificial Intelligence Supervisor: Seungryong Kim	Sep. 2024 - Present <i>Seoul, Korea</i>
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Korea University M.S./Ph.D. in Computer Science and Engineering Supervisor: Seungryong Kim	Sep. 2022 - Aug. 2024 (Incompleted) <i>Seoul, Korea</i>
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Korea University B.S. in Biomedical Engineering and Artificial Intelligence (Double Majors)	Mar. 2018 - Aug. 2022 <i>Seoul, Korea</i>
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EXPERIENCE

Meta Reality Labs Research Intern Mentors: Zhicheng Yan, Xiaoyu Xiang	Jun. 2026 - Nov. 2026 (Expected) <i>Burlingame, California</i>
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Adobe Research Research Intern Mentors: Yang Zhou, Yicong Hong, Chun-Hao Huang, Feng Liu Project: Foundation Model for Interactive Autoregressive 3D Gaming Worlds [C12]	May. 2025 - Sep. 2025 <i>San Jose, California</i>
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Google DeepMind Research Collaboration Mentors: Junhwa Hur Project: Video Diffusion Transformer for Temporal Correspondence [C9], [P1], [P3]	Dec. 2024 - Now <i>Cambridge, Massachusetts (Remote)</i>
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Adobe Research Research Intern Mentors: Yang Zhou, Zhan Xu, Jing Shi, Difan Liu, Feng Liu Project: Foundation Model for Consistent Full-Body Human Customization [C7]	May. 2024 - Aug. 2024 <i>San Jose, California</i>
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Naver Cloud Research Intern Mentors: Seunggyu Chang, Heesu Kim, DongJae Lee Project: Training-free Semantically Consistent Image Generation [C4]	Apr. 2023 - Oct. 2023 <i>Seoul, Korea</i>
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PUBLICATION

(P: Preprint, C: Conference, J: Journal, U: Under Review *: Co-first, ‡: Co-corresponding)

[P3] TrackCraft3R: Repurposing Video Diffusion Transformers for Dense 3D Tracking

Jisu Nam, Jahyeok Koo, Soowon Son, Jaewoo Jung, Honggyu An, Junhwa Hur[‡], Seungryong Kim^{*}
Arxiv, 2026

[P2] CORAL: Correspondence Alignment For Improved Virtual Try-On

Jiyoung Kim, Youngjin Shin, Siyoon Jin, Dahyun Chung, **Jisu Nam**, Tongmin Kim, Jongjae Park, Hyeonwoo Kang, Seungryong Kim
Arxiv, 2026

[P1] DiTracker: Repurposing Video Diffusion Transformers for Robust Point Tracking

Soowon Son, Honggyu An, Chaehyun Kim, Hyunah Ko, **Jisu Nam**, Dahyun Chung, Siyoon Jin, Jung Yi, Jaewon Min, Junhwa Hur, Seungryong Kim
Arxiv, 2025

[C12] WorldCam: Interactive Autoregressive 3D Gaming Worlds with Camera Pose as a Unifying Geometric Representation

Jisu Nam, Yicong Hong, Chun-Hao Paul Huang, Feng Liu, JounghBin Lee, Jiyoung Kim, Siyoon Jin, Yunsung Lee, Jaeyoon Jung, Suwhan Choi, Seungryong Kim[‡], Yang Zhou[‡]
Adobe Tech Report, CVPRW (VideoWorldModel), 2026

[C11] Deep Forcing: Training-Free Long Video Generation with Deep Sink and Participative Compression

Jung Yi, Wooseok Jang, Paul Hyunbin Cho, **Jisu Nam**, Heeji Yoon, Seungryong Kim
International Conference on Machine Learning (ICML), 2026

[C10] MATRIX: Mask Track Alignment for Interaction-Aware Video Generation

Siyoon Jin, Seongchan Kim, Dahyun Chung, Jaeho Lee, Hyunwook Choi, **Jisu Nam**, Jiyoung Kim, Seungryong Kim
International Conference on Learning Representations (ICLR), 2026

[C9] Emergent Temporal Correspondences from Video Diffusion Transformers

Jisu Nam^{*}, Soowon Son^{*}, Dahyun Chung, Jiyoung Kim, Siyoon Jin, Junhwa Hur[‡], Seungryong Kim[‡]
Neural Information Processing Systems (NeurIPS), 2025

[C8] Appearance Matching Adapter for Exemplar-based Semantic Image Synthesis

Siyoon Jin, **Jisu Nam**, Jiyoung Kim, Dahyun Chung, Yeong-Seok Kim, Joonhyung Park, HeonJeong Chu, Seungryong Kim
International Conference on Computer Vision (ICCV), 2025

[C7] Visual Persona: Foundation Model for Full-Body Human Customization

Jisu Nam, Soowon Son, Zhan Xu, Jing Shi, Difan Liu, Feng Liu, Seungryong Kim[‡], Yang Zhou[‡]
Conference on Computer Vision Pattern Recognition (CVPR), 2025

[J1] DiffFace: Diffusion-based Face Swapping with Facial Guidance

Kihong Kim^{*}, Yunho Kim^{*}, Seokju Cho, Junyoung Seo, **Jisu Nam**, Kychul Lee, Seungryong Kim[‡], KwangHee Lee[‡]
Pattern Recognition (PR), 2025

[C6] MoDiTalker: Motion-Disentangled Diffusion Model for High-Fidelity Talking Head Generation

Seyeon Kim^{*}, Siyoon Jin^{*}, Jihye Park^{*}, Kihong Kim, Jiyoung Kim, **Jisu Nam**, Seungryong Kim
Association for the Advancement of Artificial Intelligence (AAAI), 2025

[C5] Local All-Pair Correspondence for Point Tracking

Seokju Cho, Jiahui Huang, **Jisu Nam**, Honggyu An, Seungryong Kim[‡], Joon-Young Lee[‡]
European Conference on Computer Vision (ECCV), 2024

[C4] DreamMatcher: Appearance Matching Self-Attention for Semantically-Consistent Text-to-Image Customization

Jisu Nam, Heesu Kim, DongJae Lee, Siyoon Jin, Seungryong Kim[‡], Seunggyu Chang[‡]

Conference on Computer Vision Pattern Recognition (CVPR), 2024

[C3] Diffusion Model for Dense Matching

Jisu Nam, Gyuseong Lee, Sunwoo Kim, Hyeonsu Kim, Hyounghwon Cho, Seyeon Kim, Seungryong Kim

International Conference on Learning Representations (ICLR), 2024 (Oral, 1.2% acceptance rate)

[C2] Neural Matching Fields: Implicit Representation of Matching Fields for Image Matching

Sunghwan Hong, Jisu Nam, Seokju Cho, Susung Hong, Sangryul Jeon, Dongbo Min, Seungryong Kim

Neural Information Processing Systems (NeurIPS), 2022

[C1] Cost Aggregation with 4D Convolutional Swin Transformer for Few-Shot Segmentation

Sunghwan Hong*, Seokju Cho*, Jisu Nam, Stephen Lin, Seungryong Kim

European Conference on Computer Vision (ECCV), 2022

PROJECT

Kakao

Research Intern

Project: Efficient Video Generation with Latent Compression

Mar. 2026 - Jun. 2026

Seoul, Korea

Hyundai Mobis

Research Collaboration

Project: Correspondence-aware Exemplar-based Semantic Image Synthesis [C8]

Apr. 2024 - Dec. 2024

Seoul, Korea (Remote)

HONORS

37th Workshop on Image Processing and Image Understanding, KIBME

Best Paper Award

2025

37th Workshop on Image Processing and Image Understanding, KIBME

Best Poster Presentation Award

2025

Qualcomm Innovation Fellowship 2024, Qualcomm

Winner

2024

33th Artificial Intelligence and Signal Processing, IEIE

Best Paper Award

2023

CONFERENCE REVIEWER

Conference on Computer Vision Pattern Recognition (CVPR), 2023, 2024, 2025, 2026

International Conference on Learning Representations (ICLR), 2025

Neural Information Processing Systems (NeurIPS), 2026

International Conference on Computer Vision (ICCV), 2025

European Conference on Computer Vision (ECCV), 2024, 2026

International Conference on Machine Learning (ICML), 2026

Association for the Advancement of Artificial Intelligence (AAAI), 2026

Association of Computing Machinery annual conference on Multi-Media (ACM MM), 2025

Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH), 2026